

Tableau-TabPy Integration: Tableau Visualization on Risk Prediction using ML Model Deployment in TabPy Server

Created by: Mohammed Golam Kaisar Hossain Bhuyan

Date: NOV 29, 2025

◆ Go To Tableau Configuration -> Manage Analytics Extensions Connection

```
In [ ]: Hostname: localhost  
Port: 9004
```

◆ Step 1 — Install and Start TabPy

```
In [ ]: ## From Command Prompt Run Below:  
#Command 1: pip install tabpy  
#Command 2: tabpy  
## Successful run will show: TabPy server running (http://localhost:9004 by
```

◆ Step 2 — Load Your Model (.pkl) in Python

```
In [ ]: import pickle  
  
# Provide the directory path of the saved Model  
MODEL_PATH = "model/diabetes-risk-pred-kaisar-v1.1.pkl"  
  
with open(MODEL_PATH, "rb") as f:  
    model = pickle.load(f)  
  
model
```

◆ Step 3 — Connect to TabPy from Notebook

```
In [2]: from tabpy.tabpy_tools.client import Client  
client = Client("http://localhost:9004/")
```

◆ Step 4 — Define a Prediction Function for TabPy as the

```
In [3]: def predict_diabetes(a1c, glucose_pp, glucose_fast, family_history,  
                             age, activity, bmi, systolic_bp):  
  
    import pandas as pd
```

```

import numpy as np

df = pd.DataFrame({
    "hba1c": a1c,
    "glucose_postprandial": glucose_pp,
    "glucose_fasting": glucose_fast,
    "family_history_diabetes": family_history,
    "age": age,
    "physical_activity_minutes_per_week": activity,
    "bmi": bmi,
    "systolic_bp": systolic_bp
})

# Normalize family history
df["family_history_diabetes"] = (
    df["family_history_diabetes"]
    .replace({"Yes":1,"No":0,"yes":1,"no":0,True:1,False:0})
    .fillna(0)
    .astype(int)
)

# Fill missing numeric values
df = df.fillna(df.median(numeric_only=True))

# Predict
probabilities = model.predict_proba(df)[:,-1]
predicted_class = ["Yes" if p >= 0.5 else "No" for p in probabilities]

return {
    "preds": predicted_class,
    "probs": probabilities.tolist()
}

```

◆ Step 5 — Deploy Function to TabPy

```

In [4]: # Deploy to TabPy (name in Tableau as "predict_diabetes")
client.deploy('predict_diabetes',
             predict_diabetes,
             'Predicts diabetes Yes/No and probability (probs)')
print("Deployed predict_diabetes to TabPy")

```

Deployed predict_diabetes to TabPy

◆ Step 6 — Test Call Locally from Python

```

In [5]: client.query(
        'predict_diabetes',
        [2.8], [120], [100], ["Yes"], [45], [40], [27.6], [138]
    )

```

```

Out[5]: {'response': {'preds': ['No'], 'probs': [0.35]},
        'version': 1,
        'model': 'predict_diabetes',
        'uuid': 'eebf62dd-9862-494b-af22-cce733d78988'}

```

◆ Tableau Calculated Fields

◆ Tableau Calculated Field: Predicted Class

Returns the Diabetes Probability Class (Yes/No): Yes means Diabetic No Means Non Diabetic

```
In [ ]: SCRIPT_STR(
"
result = tabpy.query(
    'predict_diabetes',
    _arg1, _arg2, _arg3, _arg4,
    _arg5, _arg6, _arg7, _arg8
)['response']

return result['preds']
",
ATTR([Hba1C]),
ATTR([Glucose Postprandial]),
ATTR([Glucose Fasting]),
ATTR([Family History (Yes/No)]),
ATTR([Age]),
ATTR([Physical Activity Minutes Per Week]),
ATTR([Bmi]),
ATTR([Systolic Bp])
)
```

◆ Tableau Calculated Field: Predicted Probability

```
In [ ]: SCRIPT_REAL(
"
result = tabpy.query(
    'predict_diabetes',
    _arg1, _arg2, _arg3, _arg4,
    _arg5, _arg6, _arg7, _arg8
)['response']

return result['probs']
",
ATTR([Hba1C]),
ATTR([Glucose Postprandial]),
ATTR([Glucose Fasting]),
ATTR([Family History (Yes/No)]),
ATTR([Age]),
ATTR([Physical Activity Minutes Per Week]),
ATTR([Bmi]),
ATTR([Systolic Bp])
)
```

◆ Tableau Calculated Field: Predicted Class (Individual)

Returns the Diabetes Probability Class (Yes/No): Yes means Diabetic No Means Non Diabetic

```
In [ ]: SCRIPT_STR(
"
result = tabpy.query(
    'predict_diabetes',
    _arg1, _arg2, _arg3, _arg4,
    _arg5, _arg6, _arg7, _arg8
)['response']

return result['preds']
",
[Parameter Hba1C],
[Parameter Glucose Postprandial],
[Parameter Glucose Fasting],
[Parameter Family History],
[Parameter Age],
[Parameter Physical Activity Min per Week],
[Parameter BMI],
[Parameter Systolic BP]
)
```

◆ Tableau Calculated Field: Predicted Probability (Individual)

```
In [ ]: SCRIPT_REAL(
"
result = tabpy.query(
    'predict_diabetes',
    _arg1, _arg2, _arg3, _arg4,
    _arg5, _arg6, _arg7, _arg8
)['response']

return result['probs']
",
ATTR([Parameter Hba1C]),
ATTR([Parameter Glucose Postprandial]),
ATTR([Parameter Glucose Fasting]),
ATTR([Parameter Family History]),
ATTR([Parameter Age]),
ATTR([Parameter Physical Activity Min per Week]),
ATTR([Parameter BMI]),
ATTR([Parameter Systolic BP])
)
```

◆ Tableau Calculated Field: Probability %

```
In [ ]: [Predicted Probability (Individual)] * 100
```

◆ Tableau Calculated Field: Probability Gauge (Individual)

```
In [ ]: [Predicted Probability (Individual)] * 100
```

◆ Tableau Calculated Field: Risk Summary

```
In [ ]: "Prediction: " + [Predicted Class (Individual)]  
+ CHAR(10)  
+ "Probability: " +  
STR(ROUND([Predicted Probability (Individual)]*100,1)) + "%"
```

◆ Tableau Calculated Field: Gauge Value (numeric)

```
In [ ]: ROUND([Predicted Probability (Individual)]*100)
```

◆ Tableau Calculated Field: Gauge Remainder

```
In [ ]: 100 - [Gauge Value (numeric)]
```

◆ Tableau Calculated Field: Color Indicator

```
In [ ]: IF [Predicted Class (Individual)] = "Yes" THEN "High Risk"  
ELSE "Low Risk"  
END
```

End